

NAME: Carbonel, Nicole D.G.	Date: May 05, 2026
SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: - Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. - Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Packet Tracer - Troubleshoot EtherChannel

Objectives

Part 1: Examine the Physical Layer and Correct Switch Port Mode Issues

Part 2: Identify and Correct Port Channel Assignment Issues

Part 3: Identify and Correct Port Channel Protocol Issues

Background

A junior technician recently configured four switches. Users are complaining that the network is running slowly, and they would like you to investigate.

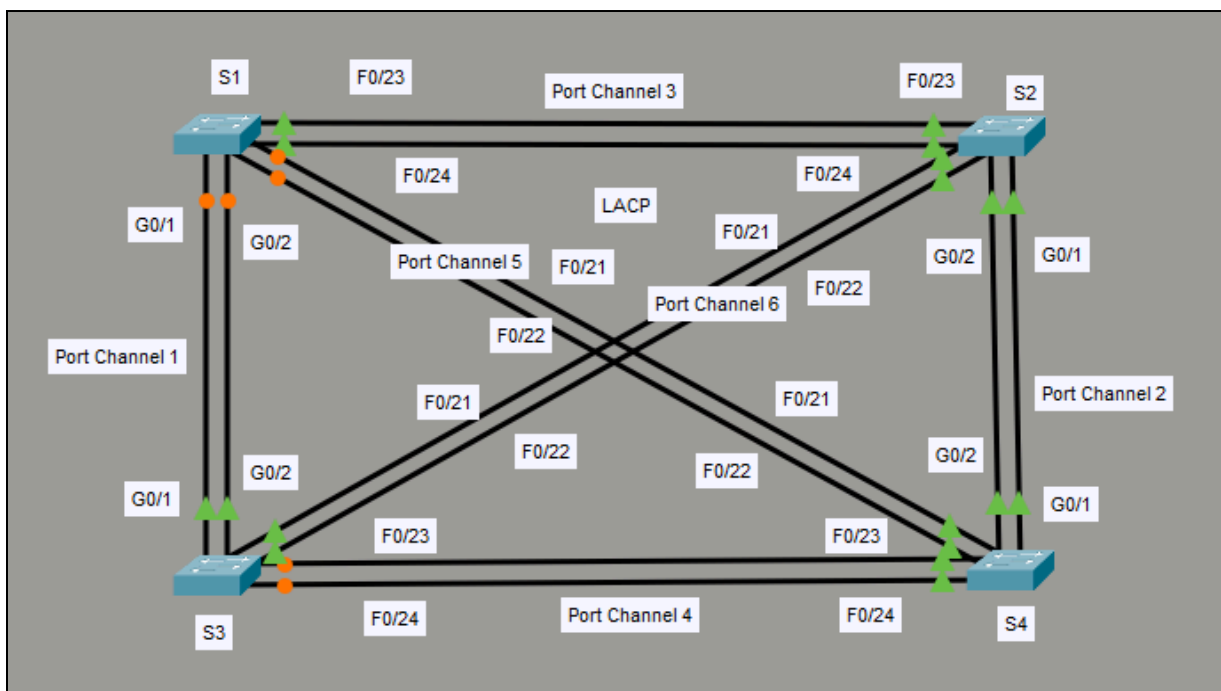
Port Channel Table

Channel Group	Ports	Protocol
1	S1: G0/1, G0/2 S2: G0/1, G0/2	LACP active
2	S2: G0/1, G0/2 S4: G0/1, G0/2	LACP active
3	S1: F0/23, F0/24 S2: F0/23, F0/24	LACP active
4	S3: F0/23, F0/24 S4: F0/23, F0/24	LACP active
5	S1: F0/21, F0/22	LACP active

Channel Group	Ports	Protocol
	S4: F0/21, F0/22	
6	S2: F0/21, F0/22 S3: F0/21, F0/22	LACP active

Device Table

Device	Group	Ports
S1	1	G0/1, G0/2
	3	F0/23, F0/24
	5	F0/21, F0/22
S2	2	G0/1, G0/2
	3	F0/23, F0/24
	6	F0/21, F0/22
S3	1	G0/1, G0/2
	4	F0/23, F0/24
	6	F0/21, F0/22
S4	2	G0/1, G0/2
	4	F0/23, F0/24
	5	F0/21, F0/22



Instructions

Part 1: Examine the Physical Layer and Correct Switch Port Mode Issues

Step 1: Look for access ports.

Examine the switches. When two or more redundant links connect the same switches, Spanning Tree protocol will only put one port in forwarding mode to prevent switching loops. You can see this in Packet Tracer. When physical ports are assigned to an EtherChannel port, they behave as one logical port. Each pair will either be operational or down.

```
S1>ena
S1#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/21    on        802.1q         trunking      1
Fa0/22    on        802.1q         trunking      1
Fa0/23    on        802.1q         trunking      1
Fa0/24    on        802.1q         trunking      1
Gig0/1    on        802.1q         trunking      1
Gig0/2    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/21    1-1005
Fa0/22    1-1005
Fa0/23    1-1005
Fa0/24    1-1005
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Fa0/21    1
Fa0/22    1
Fa0/23    1
Fa0/24    1
Gig0/1    1
Gig0/2    1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/21    none
Fa0/22    none
Fa0/23    1
Fa0/24    none
Gig0/1    none
Gig0/2    none
```

```
S2#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/21    on        802.1q         trunking      1
Fa0/22    on        802.1q         trunking      1
Fa0/23    on        802.1q         trunking      1
Fa0/24    on        802.1q         trunking      1
Gig0/1    on        802.1q         trunking      1
Gig0/2    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/21    1-1005
Fa0/22    1-1005
Fa0/23    1-1005
Fa0/24    1-1005
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Fa0/21    1
Fa0/22    1
Fa0/23    1
Fa0/24    1
Gig0/1    1
Gig0/2    1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/21    1
Fa0/22    1
Fa0/23    1
Fa0/24    1
Gig0/1    1
Gig0/2    1
```

```
S3>ena
S3#show inter
S3#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/21    on        802.1q         trunking    1
Fa0/22    on        802.1q         trunking    1
Fa0/23    on        802.1q         trunking    1
Fa0/24    on        802.1q         trunking    1
Gig0/1    on        802.1q         trunking    1
Gig0/2    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/21    1-1005
Fa0/22    1-1005
Fa0/23    1-1005
Fa0/24    1-1005
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Fa0/21    1
Fa0/22    1
Fa0/23    1
Fa0/24    1
Gig0/1    1
Gig0/2    1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/21    1
Fa0/22    none
Fa0/23    none
Fa0/24    none
Gig0/1    1
Gig0/2    1
```

```
S4>ena
S4#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/21    on        802.1q         trunking    1
Fa0/22    on        802.1q         trunking    1
Fa0/23    on        802.1q         trunking    1
Fa0/24    on        802.1q         trunking    1
Gig0/1    on        802.1q         trunking    1
Gig0/2    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/21    1-1005
Fa0/22    1-1005
Fa0/23    1-1005
Fa0/24    1-1005
Gig0/1    1-1005
Gig0/2    1-1005

Port      Vlans allowed and active in management domain
Fa0/21    1
Fa0/22    1
Fa0/23    1
Fa0/24    1
Gig0/1    1
Gig0/2    1

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/21    1
Fa0/22    1
Fa0/23    1
Fa0/24    1
Gig0/1    1
Gig0/2    none
```

Step 2: Verify ports are in trunk mode.

- Verify that all physical ports in the topology are configured as trunks. Correct any ports that are in access mode.
- Correct any EtherChannel ports that are not configured as trunks.

```
S1#config t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#
S1(config)#interface range f0/21-24, g0/1-2
S1(config-if-range)#switch
S1(config-if-range)#switchport mode trunk
S1(config-if-range)#exit
S1(config)#
S1(config)#interface port-channel 1
S1(config-if)#interface port-channel 1
S1(config-if)#exit
S1(config)#interface port-ch
S1(config)#interface port-channel 3
S1(config-if)#switchport mode trunk
S1(config-if)#exit
S1(config)#interface port-channel 5
S1(config-if)#switchport mode trunk
S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
S2>ena
S2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface range f0/21 - 24, g0/1 - 2
S2(config-if-range)#swit
S2(config-if-range)#switchport mode trunk
S2(config-if-range)#exit
S2(config)#interface port-channel 2
S2(config-if)#switchport mode trunk
S2(config-if)#exit
S2(config)#interface port-channel 3
S2(config-if)#switchport mode trunk
S2(config-if)#exit
S2(config)#interface port-channel 6
S2(config-if)#switchport mode trunk
S2(config-if)#exit
S2(config)#
```

```
S3>
S3>ena
S3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#interface range g0/1 - 2, f0/21 - 24
S3(config-if-range)#swit
S3(config-if-range)#switchport mode trunk
S3(config-if-range)#exit
S3(config)#
S3(config)#interface port-channel 1
S3(config-if)#switchport mode trunk
S3(config-if)#exit
S3(config)#interface port-channel 4
S3(config-if)#switchport mode trunk
S3(config-if)#exit
S3(config)#interface port-channel 6
S3(config-if)#switchport mode trunk
S3(config-if)#exit
S3(config)#
```

```
S4>ena
S4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S4(config)#interface range g0/1 - 2, f0/21 - 24
S4(config-if-range)#swit
S4(config-if-range)#switchport mode trunk
S4(config-if-range)#exit
S4(config)#
S4(config)#interface port-channel 2
S4(config-if)#switchport m
S4(config-if)#switchport mode trunk
S4(config-if)#interface port-channel 4
S4(config-if)#switchport mode trunk
S4(config-if)#interface port-channel 5
S4(config-if)#switch
S4(config-if)#switchport mode trunk
S4(config-if)#
S4(config-if)#exit
S4(config-if)#
```

Part 2: Identify and Correct Port Channel Assignment Issues

Step 1: Examine port channel assignments.

The Packet Tracer topology and the Port Channel and Device tables provide details about the physical ports and their EtherChannel assignments. Use the **show etherchannel summary** command to learn how the EtherChannel links are configured. Verify that the switches are configured as shown in the documentation.

```
S1#show ethercha
S1#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SD)         -           -
3      Po3 (SD)         -           -
5      Po5 (SD)         -           -
S1#
```

```
S2#show ether
S2#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 3
Number of aggregators:          3
```

Group	Port-channel	Protocol	Ports
2	Po2(SD)	-	
3	Po3(SD)	-	
6	Po6(SD)	-	

```
S3#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 3
Number of aggregators:          3
```

Group	Port-channel	Protocol	Ports
1	Po1(SD)	-	
4	Po4(SD)	-	
6	Po6(SD)	-	

```
S4#show et
S4#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 3
Number of aggregators:          3
```

Group	Port-channel	Protocol	Ports
2	Po2(SD)	-	
4	Po4(SD)	-	
5	Po5(SD)	-	

S4#

Step 2: Correct port channel assignments.

Correct any switch ports that are not assigned to the correct EtherChannel port.

```
S1#ena
S1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#interface range g0/1-2
S1(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

S1(config-if-range)#exit
S1(config)#interface range f0/23-24
S1(config-if-range)#channel-group 3 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

S1(config-if-range)#exit
S1(config)#interface range f0/21-22
S1(config-if-range)#channel-group 5 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
```

```
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface range g0/1-2
S2(config-if-range)#channel-group 2 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

S2(config-if-range)#exit
S2(config)#interface range f0/23-24
S2(config-if-range)#channel-group 3 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface Port-channel3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to up

S2(config-if-range)#exit
S2(config)#interface range f0/21-22
S2(config-if-range)#channel-group 6 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up

S2(config-if-range)#end
```



```

S3>ena
S3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#interface range g0/1-2
S3(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S3(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

S3(config-if-range)#exit
S3(config)#interface range f0/23-24
S3(config-if-range)#channel-group 4 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S3(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

S3(config-if-range)#exit
S3(config)#
S3(config)#interface range f0/21-22
S3(config-if-range)#channel-group 6 mode active
^
% Invalid input detected at '^' marker.

S3(config-if-range)#channel-group 6 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
S3(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up

%LINK-5-CHANGED: Interface Port-channel6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel6, changed state to up

S3(config-if-range)#end
S3#

```

```
S4>ena
S4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S4(config)#interface range g0/1-2
S4(config-if-range)#channel-group 2 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

S4(config-if-range)#exit
S4(config)#interface range f0/23-24
S4(config-if-range)#channel-group 4 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface Port-channel4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel4, changed state to up

S4(config-if-range)#exit
S4(config)#interface range f0/21-22
S4(config-if-range)#channel-group 5 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
S4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up

%LINK-5-CHANGED: Interface Port-channel5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel5, changed state to up

S4(config-if-range)#end
```

Part 3: Identify and Correct Port Channel Protocol Issues

Step 1: Identify protocol issues.

In 2000, the IEEE released 802.3ad (LACP), which is an open standard version of EtherChannel. For compatibility reasons, the network design team chose to use LACP across the network. The design team has made it a requirement that all ports that participate in EtherChannel need to actively negotiate the link as LACP. Verify that the physical ports are configured as indicated in the topology and Port Channel Table.

```
S1#show ether
S1#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 3
Number of aggregators:          3
```

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gig0/1(P) Gig0/2(P)
3	Po3(SU)	LACP	Fa0/23(P) Fa0/24(P)
5	Po5(SU)	LACP	Fa0/21(P) Fa0/22(P)

```
S2#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 3
Number of aggregators:          3
```

Group	Port-channel	Protocol	Ports
2	Po2(SU)	LACP	Gig0/1(P) Gig0/2(P)
3	Po3(SU)	LACP	Fa0/23(P) Fa0/24(P)
6	Po6(SU)	LACP	Fa0/21(P) Fa0/22(P)

```
S3#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port
```

```
Number of channel-groups in use: 3
Number of aggregators:          3
```

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gig0/1(P) Gig0/2(P)
4	Po4(SU)	LACP	Fa0/23(P) Fa0/24(P)
6	Po6(SU)	LACP	Fa0/21(P) Fa0/22(P)

```
S4#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
2      Po2 (SU)      LACP       Gig0/1 (P) Gig0/2 (P)
4      Po4 (SU)      LACP       Fa0/23 (P) Fa0/24 (P)
5      Po5 (SU)      LACP       Fa0/21 (P) Fa0/22 (P)
S4#
```

Step 2: Correct Protocol issues.

- Correct any switch ports that are not negotiating using LACP.
- Reissue the **show etherchannel summary** command to verify that all EtherChannel links are now correctly configured.

```

Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#interface range g0/1-2
S1(config-if-range)#no channel-group
S1(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

S1(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

S1(config-if-range)#exit
S1(config)#interface range f0/23-24
S1(config-if-range)#no channel-group
S1(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to down

S1(config-if-range)#channel-group 3 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface Port-channel3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to up

S1(config-if-range)#exit
S1(config)#interface range f0/21-22
S1(config-if-range)#no channel-group
S1(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel5, changed state to down

S1(config-if-range)#channel-group 5 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
S1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down

```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up
%LINK-5-CHANGED: Interface Port-channel5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel5, changed state to up

S1(config-if-range)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show ether
S1#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)          LACP       Gig0/1(P) Gig0/2(P)
3      Po3(SU)          LACP       Fa0/23(P) Fa0/24(P)
5      Po5(SU)          LACP       Fa0/21(P) Fa0/22(P)
S1#
```

```

Enter configuration commands, one per line. End with CNTL/Z.
S2(config)#interface range g0/1-2
S2(config-if-range)#no channel-group
S2(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down

S2(config-if-range)#channel-group 2 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

S2(config-if-range)#exit
S2(config)#
S2(config)#interface range f0/23-24
S2(config-if-range)#no channel-group
S2(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to down

S2(config-if-range)#channel-group 3 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface Port-channel3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to up

S2(config-if-range)#exit
S2(config)#
S2(config)#interface range f0/21-22
S2(config-if-range)#no channel-group
S2(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel6, changed state to down

S2(config-if-range)#channel-group 6 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down

```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up
%LINK-5-CHANGED: Interface Port-channel6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel6, changed state to up

S2(config-if-range)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console

S2#show etherch
S2#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
2      Po2(SU)        LACP       Gig0/1(P) Gig0/2(P)
3      Po3(SU)        LACP       Fa0/23(P) Fa0/24(P)
6      Po6(SU)        LACP       Fa0/21(P) Fa0/22(P)
S2#
```



```
S3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S3(config)#interface range g0/1-2
S3(config-if-range)#no channel-group
S3(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

S3(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S3(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

S3(config-if-range)#exit
S3(config)#
S3(config)#interface range f0/23-24
S3(config-if-range)#no channel-group
S3(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel4, changed state to down

S3(config-if-range)#channel-group 4 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S3(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINK-5-CHANGED: Interface Port-channel4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel4, changed state to up

S3(config-if-range)#exit
S3(config)#interface range f0/21-22
S3(config-if-range)#no channel-group
S3(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel6, changed state to down

S3(config-if-range)#channel-group 6 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
S3(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up
%LINK-5-CHANGED: Interface Port-channel6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel6, changed state to up

S3(config-if-range)#end
S3#
%SYS-5-CONFIG_I: Configured from console by console

S3#show ether
S3#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP       Gig0/1(P) Gig0/2(P)
4      Po4(SU)        LACP       Fa0/23(P) Fa0/24(P)
6      Po6(SU)        LACP       Fa0/21(P) Fa0/22(P)
S3#
```

```

S4>ena
S4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
S4(config)#interface range g0/1-2
S4(config-if-range)#no channel-group
S4(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down

S4(config-if-range)#channel-group 2 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
S4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

S4(config-if-range)#exit
S4(config)#
S4(config)#interface range f0/23-24
S4(config-if-range)#no channel-group
S4(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel4, changed state to down

S4(config-if-range)#channel-group 4 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
S4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface Port-channel4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel4, changed state to up

S4(config-if-range)#exit
S4(config)#interface range f0/21-22
S4(config-if-range)#no channel-group
S4(config-if-range)#
%LINK-3-UPDOWN: Interface Port-channel5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel5, changed state to down

S4(config-if-range)#channel-group 5 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
S4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down

```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up
%LINK-5-CHANGED: Interface Port-channel5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel5, changed state to up

S4(config-if-range)#end
S4#
%SYS-5-CONFIG_I: Configured from console by console

S4#show ether
S4#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 3
Number of aggregators:          3

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
2      Po2(SU)        LACP       Gig0/1(P) Gig0/2(P)
4      Po4(SU)        LACP       Fa0/23(P) Fa0/24(P)
5      Po5(SU)        LACP       Fa0/21(P) Fa0/22(P)
S4#
```